DM

# Assignment - 03

02/11/23

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PAGE NO.: 17

- PART - A ($2 \times 5 = 10$)

Q A1 Define group and semi-group.

Ans Group: Let S be a non-empty set and * be a binary operation then algebraic structure $(S, *)$ is said to be group if satisfy the following condition :-

i) Closure property :- $\forall a, b \in S \Rightarrow a*b \in S$.

ii) Associative property :- $\forall a, b, c, \in S \Rightarrow (a*b)*c = a*(b*c)$

iii) Existence of inverse element : $\forall a \in G$, $\exists$ an element $a' \in G$ : $a*a' = a'*a = e$
$a'$ is called the inverse element of $a$.

iv) Existence of identity element :- there exist an element $e \in S$ such that $a*e = e*a \ \forall a \in S$.

- Semi-group :- Let S be a non-empty set and * be a binary operation then algebraic structure $(S, *)$ is said to be semi-group if satisfies the following condition :-

i) closure property

ii) Associative property.

Q2. The rational numbers Q defined by $a*b = ab/4$ for all $a, b, \in Q$. Determine whether operation * is (a) commutative (b) associative.

Sol. $a*b = ab/4$ (* is commutative if)

$a*b = b*a$

$a*b = \frac{ab}{4}$

$b*a = \frac{ba}{4} = \frac{ab}{4}$

$\Rightarrow a*b = b*a$ $\therefore$ * is commutative.

b) $*$ is associative if

$(a*b)*c = a*(b*c)$

$(a*b)*c$

$= \left(\frac{ab}{4}\right)*c$

$\frac{\left(\frac{ab}{4}\right)*c}{4}$

$= \frac{abc}{16}$

$a*(b*c)$

$= a*\left(\frac{bc}{4}\right)$

$= \frac{a*\left(\frac{bc}{4}\right)}{4}$

$= \frac{abc}{16}$

$\Rightarrow (a*b)*c = a*(b*c)$

$\therefore *$ is associative.

Q3. Define conjunction and disjunction with truth tables.

Ans Conjunction AND ($\wedge$): Let p & q are 2 preposition the conjunction of p & q is the preposition "p & q" written $p \wedge q$, that is true when both p & q are true and false otherwise.

Truth table :-

| p | q | $p \wedge q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | F |

- Disjunction (OR)($\vee$):- Let p & q be two prepositions. The disjunction of p & q is preposition "p or q" written $p \vee q$ that is true when atleast one of the 2 preposition are true.

Truth table :

| p | q | $p \vee q$ |
|---|---|---|
| T | T | T |
| T | F | T |
| F | T | T |
| F | F | F |

Q4. What is conditional and biconditional statement?

Ans

- Conditional statement : Let p and q are 2 prepositions, the implication, "If p then q" written $p \rightarrow q$ is the position that it is false when p is true and q is false and true Otherwise; the implication $p \rightarrow q$, p is called hypothesis & q is called conclusion.

- Biconditional Statement : Let p & q are 2 preposition, the equivalence "p is equivalent to q" written $p \leftrightarrow q$ is the position that is true when p & q both are either true or false otherwise false.

  Symbolicaly $p \equiv q$

Q5. What is tautology, contradiction and contingency?

Ans Tautology : The compound statement that is always true for all possible truth values of its prepositional value is called tautology.

Contradiction : The compound statement that is always false is called contradiction.

Contingency : A statement that is neither tautology nor contradiction is called contingency.

PART - B

Q1. Prove that the set of rational numbers excluding zero with binary operation '$*$ multiplication' forms an abelian group.

Sol. Set of rational numbers excluding zero is denoted as $Q^*$.

(i) closure :- $a, b \in Q^* \Rightarrow a * b \in Q^*$

(ii) Associative :- Multiplication of rational numbers is associative

$\therefore a, b, c \in Q^* \Rightarrow a * (b * c) = (a * b) * c$

(iii) identity element :- identity element for multiplication is 1.

$\therefore$ for $a \in Q^* \; \exists \; a * 1 = 1 * a = a$.

(iv) inverse element : $\forall a \in Q^* \; \exists \; a^{-1} \in Q^*$

$\therefore a * a^{-1} = a^{-1} * a = 1$

$\therefore$ 1 is the inverse element.

(v) Commutative : $a, b \in Q^*, \; a * b \in Q^*$

$\Rightarrow b * a \in Q^*$

$\because Q^*$ satisfies all 5 properties

$\therefore$ it forms an abelian group under Multiplication.

Q2. Show that the propositions $p \to q$ and $(\sim p \vee q)$ are logically equivalent.

Ans Truth table for $p \to q$

| $p$ | $q$ | $p \to q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | T |
| F | F | T |

· Truth table for $(\sim p \wedge \sim q)$

| $p$ | $q$ | $\sim p$ | $\sim q$ | $\sim p \wedge \sim q$ |
|---|---|---|---|---|
| T | T | F | F | F |
| T | F | F | T | F |
| F | T | T | F | F |
| F | F | T | T | T |

$p \to q$ & $(\sim p \wedge \sim q)$ are not always same. they are only equivalent in case when p is true and q is false or both of them are false.

Q3. Show by truth table that the following statements are contradiction.

(i) $(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$

Sol. Truth table:

| $p$ | $q$ | $p \wedge q$ | $\sim p$ | $\sim q$ | $\sim p \vee \sim q$ | $(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$ |
|---|---|---|---|---|---|---|
| T | T | T | F | F | F | F |
| T | F | F | F | T | T | F |
| F | T | F | T | F | T | F |
| F | F | F | T | T | T | F |

∵ all the statement are false in each case

∴ $(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$ is contradiction.

(ii) $((p \wedge \sim q) \wedge (p \rightarrow q))$

Sol. Truth table :

| $p$ | $q$ | $\sim q$ | $p \wedge \sim q$ | $p \rightarrow q$ | $((p \wedge \sim q) \wedge (p \rightarrow q))$ |
|---|---|---|---|---|---|
| T | T | F | F | T | F |
| T | F | T | T | F | F |
| F | T | F | F | T | F |
| F | F | T | F | T | F |

$\because$ all the statement are false in each case

$\therefore$ $((p \wedge \sim q) \wedge (p \rightarrow q))$ is contradiction.

• PART - C

Q1. G is the set of rationales except $-1$, binary operation $*$ is defined by $a * b = a + b + ab$, Show that it is a group.

Sol. (i) Closure Property : Let $a, b \in G$ ; $a * b = a + b + ab \in G$

closure is satisfy.

(2) Associative : Let $a, b, c \in G$ $(a * b) * c = a * (b * c)$

$(a * b) * c = (a + b + ab) * c$

$= a + b + ab + c + (a + b + ab) c$

$= a + b + ab + c + ac + bc + abc$

$a * (b * c) = a * (b + c + bc)$

$= a + (b + c + bc) + a(b + c + bc)$

$= a + b + c + bc + ab + ac + abc$

So, $(a * b) * c = a * (b * c)$

Associative satisfied.

(3) Existence of identity : Let $a \in G$, identity element $e \in G$

$a*e = e*a = a$

$a*e = a$

$a+e+ae = a$

$e(1+a) = 1$

$\boxed{e = \frac{1}{1+a}}$ $a \neq -1$

(4) inverse :- Let $a \in G$ & inverse $(a)^{-1} \in G$ then.

$\Rightarrow a*(a^{-1}) = (a^{-1})*a = e$

$\Rightarrow a*(a^{-1}) = e$

$\Rightarrow a + a^{-1} = a(a^{-1}) = 1/1+a$

$\Rightarrow a + a^{-1} + 1 = \frac{1}{1+a}$

$\Rightarrow a^{-1} = \frac{1}{1+a} = (a+1)$

So, G is a group.

Q2. Construct truth table for each of the following propositions.

(i) $(p \vee q) \wedge (p \vee r)$

| p | q | r | $p \vee q$ | $p \vee r$ | $(p \vee q) \wedge (p \vee r)$ |
|---|---|---|---|---|---|
| T | T | T | T | T | T |
| T | F | T | T | T | T |
| T | T | F | T | T | T |
| T | F | F | T | T | T |
| F | T | T | T | T | T |
| F | F | F | F | F | F |
| F | T | F | T | F | F |
| F | F | T | F | T | F |

(ii) $\sim(p \wedge q) \wedge (\sim p \vee \sim q)$

Sol.

| p | q | ~p | ~q | p∧q | ~(p∧q) | ~p∨~q | ~(p∧q)∧(~p∨~q) |
|---|---|---|---|---|---|---|---|
| T | T | F | F | T | F | F | F |
| T | F | F | T | F | T | T | T |
| F | T | T | F | F | T | T | T |
| F | F | T | T | F | T | T | T |

Q3. Show by truth table the following formulae are tautologies.

(iii) $(p \to q) \leftrightarrow (\sim p \vee q)$

Sol.

| p | q | ~p | p → q | ~p∨q | (p→q)⟺(~p∨q) |
|---|---|---|---|---|---|
| T | T | F | T | T | T |
| T | F | F | F | F | T |
| F | T | T | T | T | T |
| F | F | T | T | T | T |

∵ all the statement is true for all possible value.
∴ $(p \to q) \leftrightarrow (\sim p \vee q)$ is a tautologies.

(iv) $((p \to q) \wedge (q \to r)) \to (p \to r)$

| p | q | r | p→q | q→r | (p→q)∧(q→r) | p→r | ((p→q)∧(q→r))→(p→r) |
|---|---|---|---|---|---|---|---|
| T | F | T | F | T | F | T | T |
| T | T | F | T | F | F | F | T |
| F | T | T | T | T | T | T | T |
| F | F | F | T | T | T | T | T |
| F | T | F | T | F | F | T | T |
| F | F | T | T | T | T | T | T |
| T | F | F | F | T | F | F | T |
| T | T | T | T | T | T | T | T |

∵ all the statement is true for all possible value.
∴ $((p \to q) \wedge (q \to r)) \to (p \to r)$ is a tuatologies.

Dec 26, 2023, 16:20